

WILMS

Women's Interest-Free Loan Management System

Product Dossier — v1.7.2

Executive Summary

WILMS v1.7.2 packages a production interest-free loan management platform for market presentation to government, NGO, and institutional partners.

Product Overview

Women's Interest-Free Loan Management System covering registration, approval, disbursement, collections, reconciliation, communications, executive intelligence, and operations.

Problem Statement

Paper and spreadsheet programmes lack separation of duties, auditability, and timely portfolio visibility.

Solution

A modular TypeScript monolith on Next.js + Vercel + Neon with RBAC, operational ledgers, notifications, and board reporting.

Target Users

Collectors, officers, approvers, auditors, super admins, directors, MPs, NGO boards, procurement committees, and investors.

Features

Borrower lifecycle, pools, collections, reconciliation, expenses, communication center, executive intelligence, exports, ops incidents.

Architecture

Next.js App Router hosts @wilms/domain via Route Handlers; custom HMAC sessions; Neon PostgreSQL; optional Redis rate limits.

BRD Summary

Interest-free group lending with maker-checker controls, field GPS collections, and programme capital hard-stops.

Implemented Modules

Core lending through v1.7.0 intelligence/exports/ops; v1.7.2 market packaging and dashboard separation.

Financial Model

Pesewas integer money; pool replenishment/disbursement/repayment/adjustment; expenses affect operating cash only.

Security Model

HMAC sessions, RBAC + overrides, CSRF on mutating BFF paths, audit logging, upload allowlists.

Operations

Health, metrics, cron notifications, incidents, maintenance windows, Neon backups.

Deployment

Vercel production + Neon; migrations via domain SQL journal.

Reporting

Operational reports, executive intelligence, forecasting, export center.

Compliance

Audit trail, SoD on expenses/adjustments, confidentiality notices on exports.

Testing

Vitest, Playwright, financial/RBAC/notification smokes, version and migration verifies.

Certification Status

Production operational platform; statutory GL and multi-org deferred.

Skipped / Deferred

Borrower portal, multi-organization tenancy, statutory GL, native mobile app, deep payment-provider integrations.

Roadmap

v1.8 Integrations & Payments; v1.9 Enterprise automation; v2.0 General Ledger & multi-branch.

Technical Appendix

Node 22, Next.js 14, Drizzle, shared-rbac, export engines (PDF/Excel/DOCX), docs/v1.7.2 release pack.